BECS 22451
Experiment No. 10

MATLAB SIMULINK FOR CONTROL

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Experiment Name: MATLAB SIMULINK FOR CONTROL

APPARATUS

- A PC or Laptop with MATLAB Simulink installed.

THEORY AND DIAGRAMS

- Control System Simulation with Simulink: Control systems play a pivotal role in regulating and influencing the behavior of dynamic systems. Simulating these systems provides a valuable means to study and analyze their responses under different conditions.
- Control Systems and Simulink: Control systems are crucial in engineering to manage and manipulate the behavior of dynamic systems. Simulink, a graphical programming environment in MATLAB, facilitates the modelling, simulation, and analysis of these systems. It offers an intuitive platform for designing and visualizing control system models.
- Second-Order System Model: A second-order system is characterized by two poles in its transfer function, representing its dynamic response. In this experiment, we will construct a second-order system model using Simulink. The model's parameters, such as damping ratio and natural frequency, significantly influence the system's behavior.
- Step Input and System Response: A step input is a fundamental test signal used to analyze the dynamic response of a system. When applied to a control system, it induces a sudden change, allowing us to observe how the system reacts over time. We can visualize and analyze the system's response to a step input through Simulink, gaining insights into its stability and transient characteristics.

Practical Scenario

Solar tracking is a technology designed to enhance the efficiency of solar energy capture by aligning solar panels or mirrors with the sun's position throughout the day. Unlike fixed solar installations, which remain in a static position, solar tracking systems dynamically adjust the orientation of solar components to maximize exposure to sunlight, as shown in Figure 1.

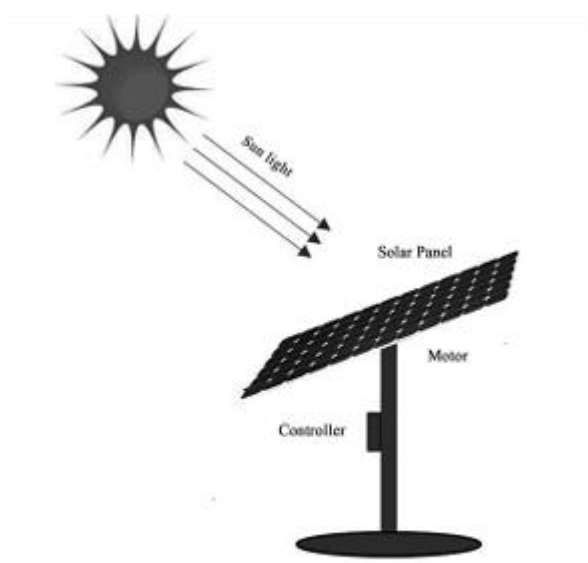


Figure 1

THEORY AND DIAGRAMS

- Motion of the Solar Panel: The rotational motion of the solar panel can be described using the second-order differential equation (1) below.

$$\frac{d^2\theta}{dt^2} = \frac{1}{J}(T - K_d \frac{d\theta}{dt}) \quad (1)$$

Where,

- $T = K_g K_t i$
- $\frac{d^2\theta}{dt^2}$: Second derivative of the tracking angle with respect to time, representing the rotational dynamics of the solar panel.
- J: Moment of inertia of the solar panel. It quantifies the distribution of mass around the axis of rotation.
- K_d: Damping constant. It represents the damping or resistive forces acting against the rotation of the solar panel.
- T: Torque of the motor
- K_t : Torque constant, relating the motor current to the torque produced.

- Electrical Dynamics of the Motor: The electrical dynamics of the motor can be described using the first-order linear differential equation (2) below.

$$\frac{di}{dt} = \frac{1}{L}(V - K_g K_f \frac{d\theta}{dt} - Ri) \quad (2)$$

- $\frac{di}{dt}$: Rate of change of current with respect to time, representing the electrical dynamics of the motor.
- L: Inductance of the motor circuit. It measures the ability of the coil to induce a voltage in itself when the current through it changes.
- V: Applied voltage to the motor.
- K_g: Gain constant. It represents the motor gain, which relates the electrical input to the mechanical output.
- K_f : Feedback constant. It accounts for any feedback mechanism influencing the motor dynamics.
- $\frac{d\theta}{dt}$: Rate of change of the tracking angle with respect to time. It's a coupling term indicating that the motor's current is influenced by the rate of change of the tracking angle.
- R: Resistance in the motor circuit.

THEORY AND DIAGRAMS

- PID Control Systems: PID (Proportional-Integral-Derivative) control optimises solar trackers by continuously adjusting panel orientation for maximum sunlight exposure. The proportional term (P) responds to current errors, integral (I) eliminates steady-state error over time, and derivative (D) anticipates future errors, preventing overshooting. This dynamic PID algorithm ensures precise tracking as the sun moves across the sky. It enhances tracking accuracy, adapts to environmental changes, and maximises energy harvesting. The result is an efficient and responsive solar tracker, crucial for optimising energy capture in solar power systems and contributing to increased overall energy efficiency.

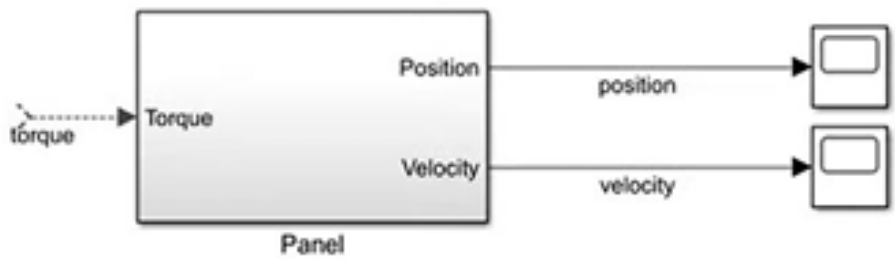


Figure 2

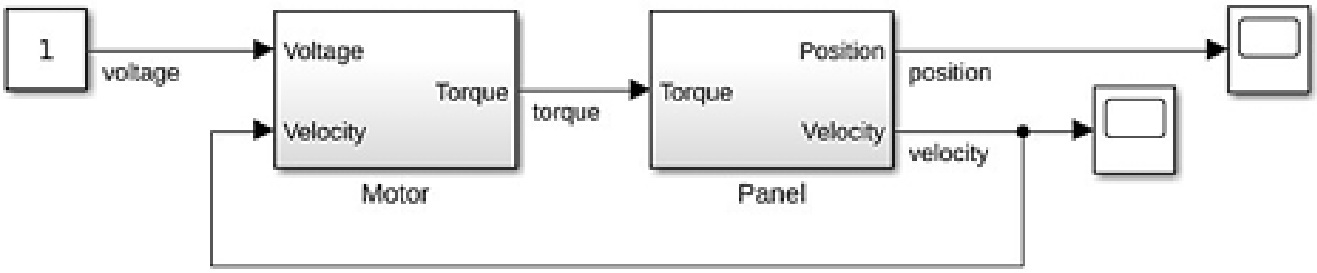


Figure 4

THEORY AND DIAGRAMS

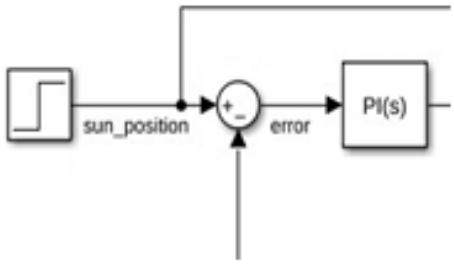


Figure 5

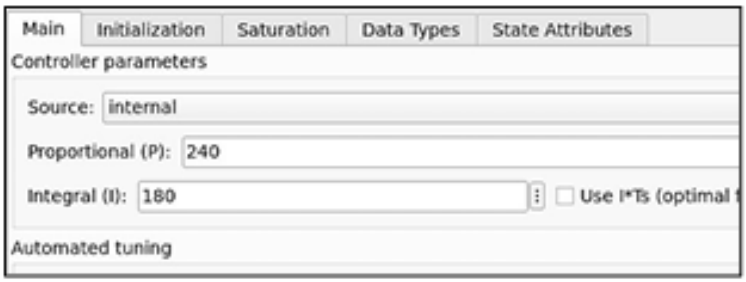


Figure 6

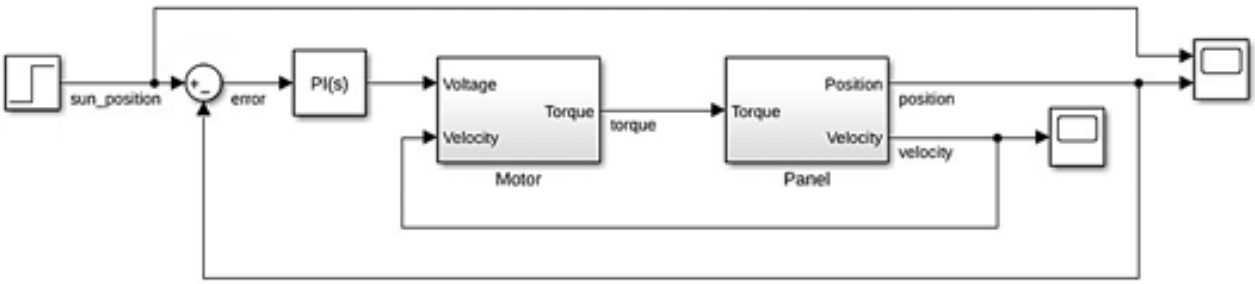


Figure 7

PROCEDURE

Part 01

1. MATLAB was started.
2. Simulink was opened, and a “Blank Module” was created in the simulation start window.
3. The Simulink project was saved as “part_01.slx” in a folder named “EC_2021_005.” A Word file was created and renamed with the student number, then saved in the same folder.
4. The Simulink model was designed to control the motion of the solar panel as shown in Figure 1, referring to equation (1). The following steps were followed:

    a. A constant block from Sources was used as input, and its value was changed to 10, representing the torque input to the system.
    b. An integrator block was used to integrate and obtain velocity and position from acceleration.
    c. The damping constant Kd was defined in a gain block by clicking the three dots and creating Kd. Its value was set to 5 and the location to Base Workspace.
    d. Similar steps as in 4(c) were followed for J, with a value of 8.6 and Base Workspace as the location.
    e. A text annotation was added using the following method of writing to display the equation (1): **\frac{d^2\theta}{dt^2}=\frac{1}{J}(T-K_d\frac{d\theta}{dt}\right)**
    f. Two scopes were attached to observe position and velocity. Components were selected, and a subsystem was created from the selection. The final model for the Motion of the Solar Panel resembled as shown in Figure 2.
    g. Outputs were observed (using 10 for simulation stop time), and screenshots were taken.

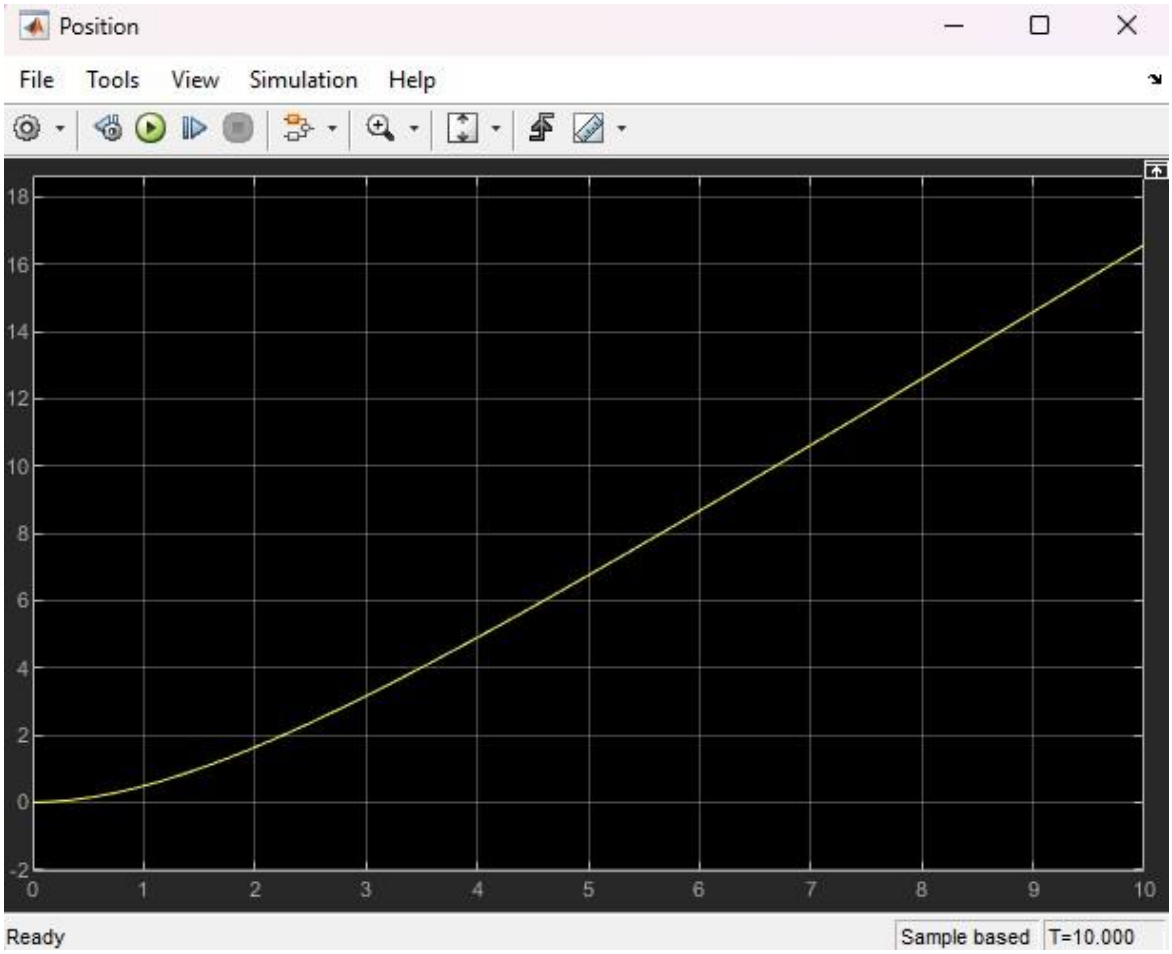
Part 02

1. The Simulink model for the electrical dynamics of the motor was designed, referring to equation (2). The following instructions were followed:
    a. Values were defined in gain blocks:
        i. L: 1.0000e-05
        ii. Kg: 2000
        iii. Kf: 0.0700
        iv. R: 10
        v. The location was set to Base Workspace.
    b. A subsystem was created using the knowledge from Part 01, resulting in a model resembling Figure 3.
    c. The model was frequently saved in the created folder.
2. A Step Function (Unit step) was inserted, a Sum block was added, and the sign list was changed to |+ . A PID controller block was added from the library browser and installed, as shown in Figure 4. A step function (Unit Step) was inserted, along with a Sum block with a |+- configuration.
3. The PID controller block was double-clicked, and values were changed as in figure 6 The controller was set to PI mode since the D term was unnecessary for slow sun movement.
4. The final model was designed as shown in Figure 5. The position scope was used to observe the sun and solar panel positions
5. The simulation was run, and the outputs were observed using scopes.

OBSERVATIONS

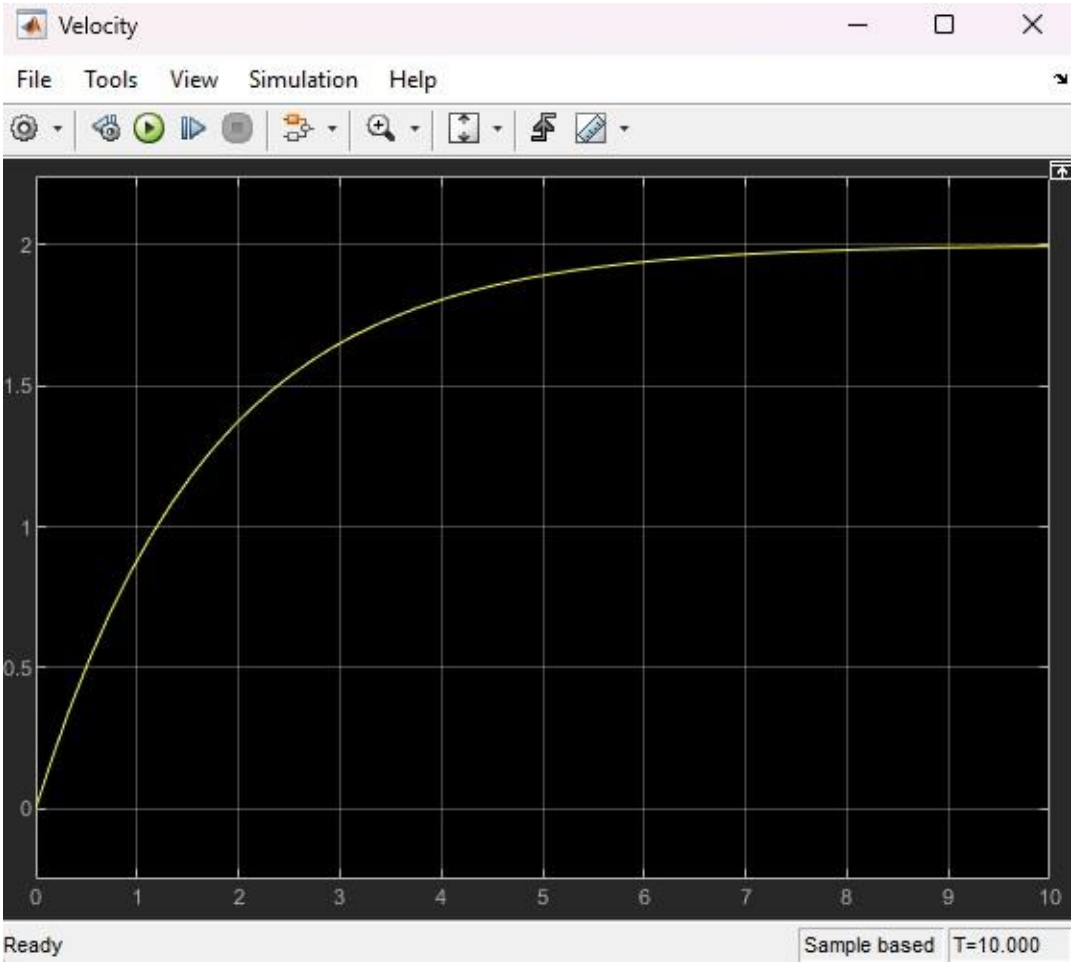
Part 01

- Graph of Position



OBSERVATIONS

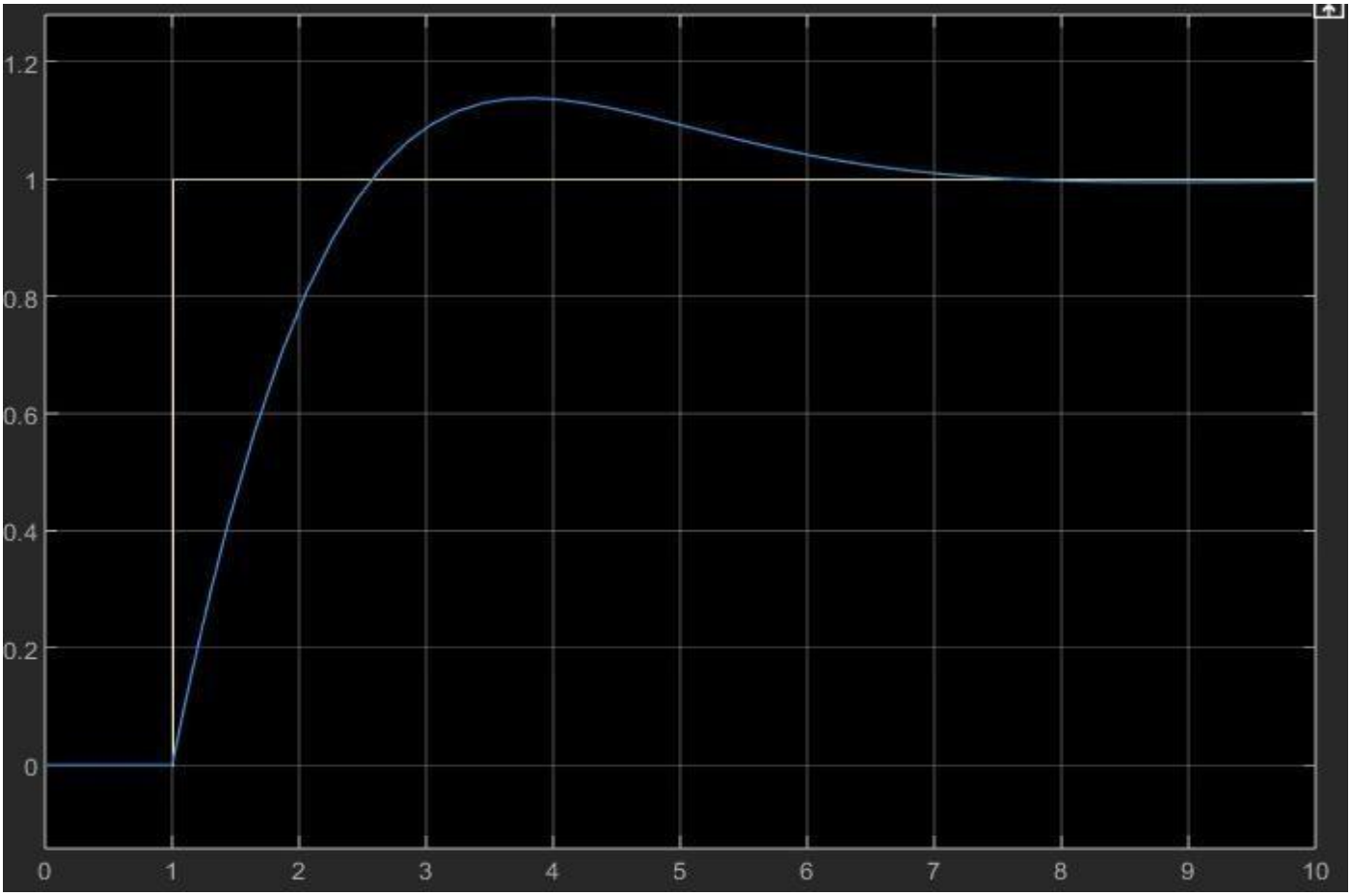
- Graph of velocity



OBSERVATIONS

Part 02

Graphs of Sun position, position



CONCLUSION

This MATLAB Simulink laboratory practical showed how modeling and simulation could be applied in the design of control systems. Second-order system construction, analysis, and PID control were beneficial in gaining insights into the system stability and response. Certain problems appeared during the experiment including setting up the parameter and tuning the PID controller; however, all these problems could be solved systematically step by step. This method provided a high level of accuracy of the obtained results and underlined the applicability of the studied theory. In general, the experiment demonstrated the importance of simulation technologies in contemporary engineering activity and provided a profound understanding of the control system construction and its practical significance.

DISCUSSION

This experiment sought to apply MATLAB Simulink in designing and simulating control systems, which are central in engineering for controlling the operation of dynamic systems. MATLAB Simulink provides a graphical user interface for the modeling and testing of these systems. The objective of the experiment was to design a second order system model for the movement of the solar panel and the electrical system of its motor. Furthermore, a PID controller was incorporated in the system to improve the performance, in which real-life application of control theory in solar tracking system was demonstrated. Some of the challenges observed during the experiment include: The first issue that was encountered with this system was setting the system parameters including the damping constant for the system and moment of inertia for the model; this was because determining how these parameters affect the model requires a lot of effort. Day by day, values were assigned which in turn caused wrong outputs and hence, correction was done repeatedly. Using Simulink to divide the model into subsystems using the "Create Subsystem" option also posed some problems because wrong connections often led to simulation errors. The most challenging activity was to set up a PID controller in which the proportional and integral gains had to be well tuned and adjusted to control the system and give the desirable response. These values could change significantly with only small differences, and system behavior was affected by these changes. Maintaining fidelity of outputs and addressing simulation errors was time consuming and attention demanding. All these challenges were tackled systematically with the help of theoretical pointers as well as practical experimenting. Theoretical equations give realistic initial conditions of system parameters, and they were adjusted in iteration for maximum performance. Subsystems were developed to alleviate the problem of large models making them manageable and easier to debug. The PID controller tuning was done gradually; basic proportional and integral gains were first applied to the system and adjustments to them were made with respect to the scopes outputs. Since the motion of the solar panel is slow and smooth, the derivative term was eliminated since oscillations were not an issue in the system. This approach made it possible to have a stable and a smooth system response.

MATLAB Simulink was found to be very useful in this experiment. Through this it made it easy to model and depict systems through its graphical user interface; other features such as scopes allowed the analysis of system's conduct in real time. This made it possible to investigate transient responses and system stability. Due to Simulink's ability to change parameters and try out different settings without having to write a lot of code, it ideal to use for learning and testing. But there are some issues in Simulink also. To the new users, the tool is complicated and learning it may take a while, which may be discouraging. When a block misconfiguration or an incorrect parameter value is passed to a function, the errors are sometimes tough to debug. Furthermore, executing simulations of large models or those with complex dynamics can be demanding in terms of system resources, resulting in slow performance on computers with limited capabilities. The position and velocity graphs were used to study the motion of the solar panel in a efficient manner. The position graph showed a gradual rise, and this could be attributed to the gradual revolution of the solar panel in response to torque applied. The velocity graph depicted a sharp rise at the onset and lesser fluctuation in the subsequent cycles implying that the system achieved equilibrium nearly immediately at the beginning of the test and fluctuated only slightly thereafter. These results demonstrate that the control system can handle the motion of the solar panel. The absence of oscillation and damping observed in both position and velocity confirm that the system is well-damped to avoid oscillation or instability.